

Brazil's Economic Development Agenda

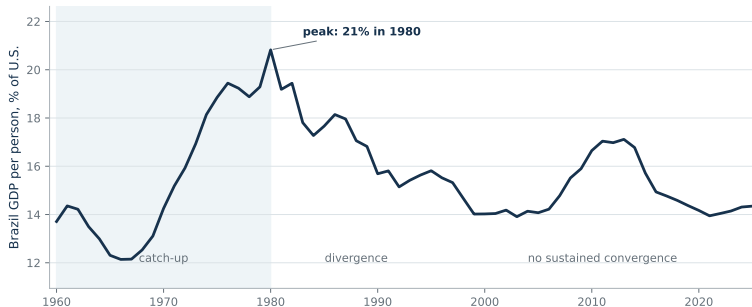
Macroeconomic stability, productivity, state capacity, and inclusion

Eduardo Peres Cunha and Arthur Lins de Vasconcellos

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Data cutoff: July 2026

Relative to U.S. income, Brazil is where it was in 1960



Brazil converged rapidly until 1980. Why did it stop, and what would make it resume?

Source: World Bank, World Development Indicators, GDP per capita in constant 2015 US\$ (NY.GDP.PCAP.KD), 1960–2025.

Development is neither linear nor irreversible

Countries can fall behind

Argentina and Venezuela show that earlier prosperity does not guarantee continued convergence.

Brazil remains in the middle

A large and diversified economy has not completed the transition to rich-country productivity and income.

Countries can move closer

Chile and South Korea illustrate different paths toward higher income and development.

Why has Brazil remained in the middle, and what would be required to complete the transition?

Source: World Bank, World Development Indicators. Country examples describe development trajectories, not formal income-group classifications.

Brazil's position raises two questions

Part I: Why are we like this?

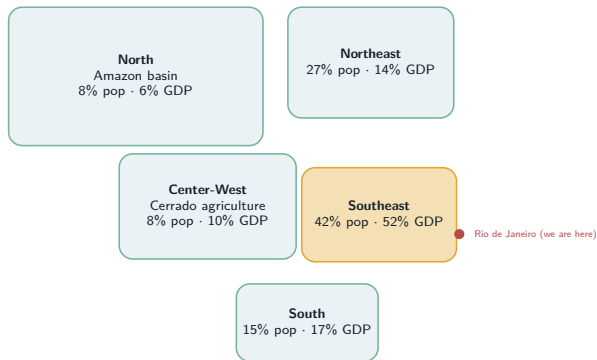
How did macroeconomic instability, slavery, rapid urbanization, and natural resources shape institutions, inequality, and production?

Part II: Where do we go from here?

Which structural problems must Brazil solve, and which productive capabilities can it expand?

The development agenda begins with Brazil's history and ends with a choice about its capabilities.

One country, five very different economies

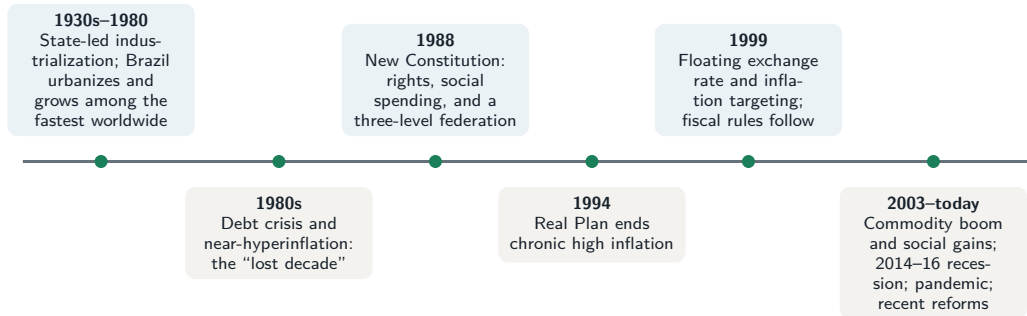


Schematic: positions indicative, areas not to scale.

- ▶ Population and output concentrate near the coast; the Southeast alone produces about half of GDP.
- ▶ The Northeast holds 27% of the people and 14% of the output — the largest regional income gap.
- ▶ The Amazon and the Cerrado make land use, agriculture, and climate part of the growth agenda.
- ▶ Keep this map in mind: the binding constraint differs by place.

Source: IBGE, 2022 Population Census and Regional Accounts 2021; shares rounded.

Six moments explain Brazil's starting point



- The lecture concentrates on the period after 1994, but the achievements and scars of this longer history set the stage.

Development is broader than current GDP growth

Income	Productivity	Inclusion	Resilience
Income per person	Output per worker	Mobility and access	Climate and institutions

$$\underbrace{\frac{Y}{N}}_{\text{income per person}} = \underbrace{\frac{Y}{L}}_{\text{output per worker}} \times \underbrace{\frac{L}{N}}_{\text{workers per person}}$$

Y = total income, L = workers, N = population

- ▶ Productivity determines how much each worker produces.
- ▶ Demography and labor-force participation determine how many people work.
- ▶ Distribution determines who can invest in skills and participate.

Six terms you will hear all afternoon

Income convergence	A poorer country grows faster than rich ones and closes the gap in income per person.
Nominal anchor	The variable a central bank commits to (an exchange rate, an inflation target) so that everyone's prices and wages can coordinate on it.
Pass-through	How much of a currency depreciation ends up in consumer prices.
Primary balance	Government revenue minus spending, <i>excluding</i> interest payments.
Reallocation	Workers and capital moving from less productive firms to more productive ones.
Contestability	Markets where new firms can credibly enter and challenge incumbents.

Everything else — URV, CBS/IBS, Selic — is defined when it appears.

The lecture follows two questions

Why are we like this?

- ▶ Historical formation
- ▶ Institutions and macroeconomic stability
- ▶ Inequality and productive structure
- ▶ Brazil today

Where do we go from here?

- ▶ Problems to solve
- ▶ Productive opportunities
- ▶ Political economy and implementation
- ▶ An integrated development agenda

Backup slides contain formulas, full tables, measurement details, and references.

Part I – Why are we like this?

History, institutions, inequality, and production

Four historical forces still shape Brazil today

Macroeconomic instability

Inflation and debt crises shortened planning horizons and weakened public investment.

Slavery

Forced labor and exclusion left persistent racial gaps in assets, education, and political power.

Rapid urbanization

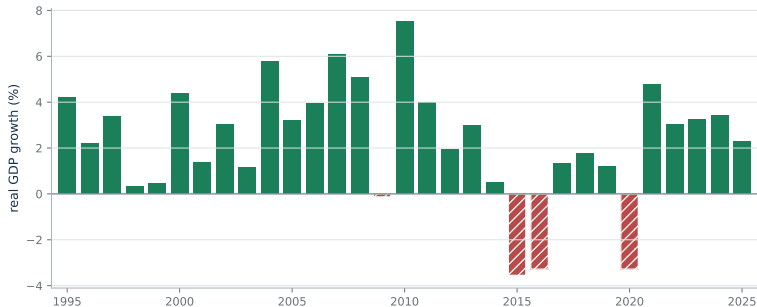
Cities expanded faster than housing, transport, sanitation, and local state capacity.

Natural resources

Land, energy, and minerals created global advantages and recurring governance conflicts.

Historical formation shaped Brazil's institutions, inequality, and productive structure.

Growth has been volatile since stabilization



- ▶ Brazil combined strong expansions with three recessions after 1995; average growth weakened before the pandemic.
- ▶ This volatility explains *why* the convergence line you saw never trends upward for long.

Source: BCB SGS 7326, based on IBGE National Accounts. Annual real GDP growth; 2025 is the latest observation. Hatched bars mark contractions.

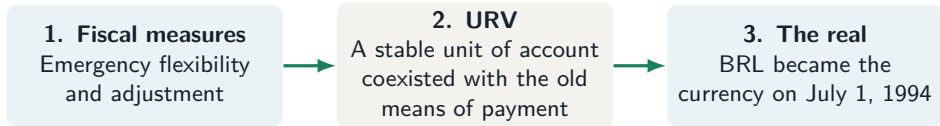
High inflation disrupts more than purchasing power



- ▶ Stabilization restored price information and lengthened planning horizons.
- ▶ The later spikes show why credibility, fiscal policy, and the exchange rate still interact.

Source: BCB SGS 433, monthly IPCA. Twelve-month rates are compounded from the official monthly series.

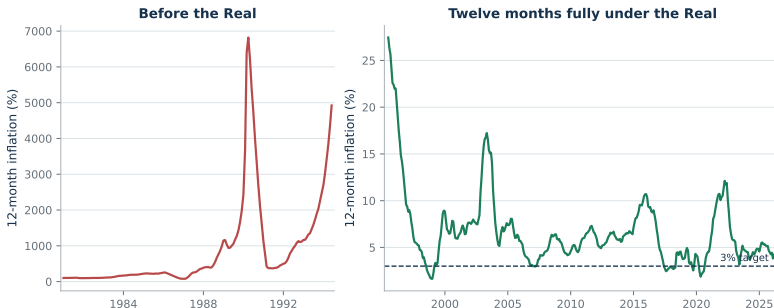
The Real Plan changed the unit of account before the currency



Concrete example: a store listed rice at **URV 5** for months while the price tag in cruzeiros reais changed almost daily with the URV's dollar-linked value. On July 1, 1994, "URV 5" simply became **R\$5**.

- The URV coordinated prices and expectations without an overnight freeze — an institutional solution, not just a new banknote.

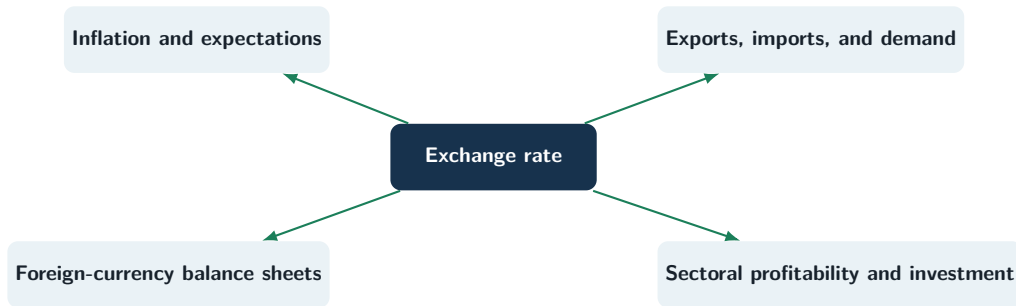
The Real Plan produced a discrete break in inflation



- ▶ Twelve-month inflation reached thousands of percent before July 1994.
- ▶ From July 1995 onward, every month in the twelve-month window uses prices quoted under the Real.

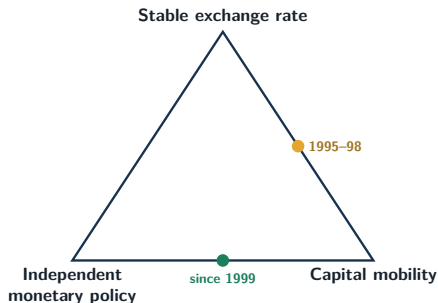
Source: BCB SGS 433, monthly IPCA. Rates are compounded over twelve months. The two panels use different vertical scales.

The exchange rate links the macroeconomy to development



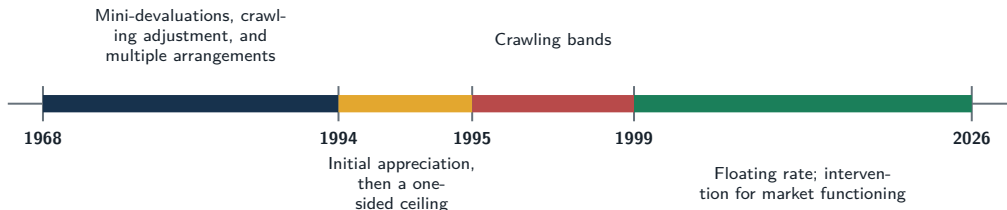
An exchange-rate regime allocates adjustment across the currency, interest rates, reserves, and domestic demand.

The policy trilemma: Brazil moved corners in 1999



- ▶ A country can fully pursue only two corners at once.
- ▶ **1995–98:** a quasi-stable exchange rate with open capital markets — so interest rates had to defend the currency.
- ▶ **Since 1999:** monetary policy targets inflation and the currency floats.
- ▶ Capital-flow measures change the constraint but do not remove it.

1994–1999: the exchange rate served as the nominal anchor

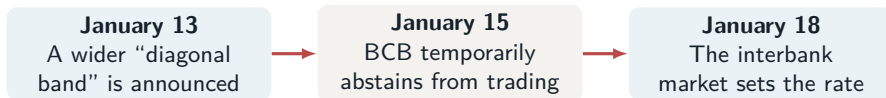


- ▶ **Why it helped:** a stable currency price disciplined tradable-goods prices and coordinated expectations after hyperinflation.
- ▶ **Why pressure accumulated:** domestic inflation exceeded foreign inflation, the real appreciated, and high interest rates defended the corridor.
- ▶ External shocks (Asia 1997, Russia 1998) repeatedly tested reserve and interest-rate defenses.

Source: BCB, 1999 FX Market Analysis and official FX-policy pages; dates before 1994 are a schematic summary.

Decision point: January 1999

It is January 12, 1999. Reserves are falling fast. You run the central bank. Do you raise interest rates again, devalue within the band, impose capital controls, or float the currency?



- ▶ None of the options was free: each allocates the adjustment to someone — borrowers, exporters, taxpayers, or savers.
- ▶ Formal inflation targeting followed in July 1999.

Source: BCB, [Communiqué 6,565](#); BCB Working Paper 25.

1999 changed who absorbs the shock

Before: defend the parity

Interest rates and reserves absorbed external shocks so the exchange-rate path could hold.

After: let the price move

The currency absorbs part of the shock; the policy rate responds to the inflation outlook.

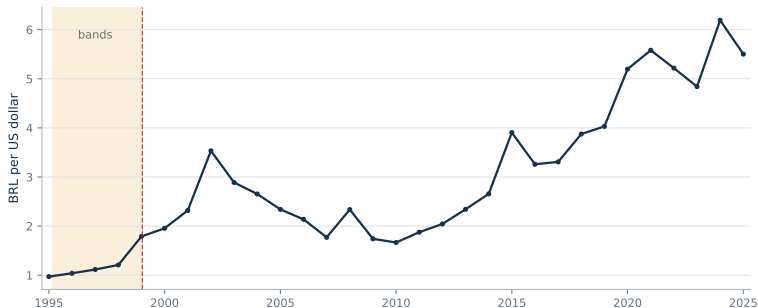
Adjustment since 1999

The market determines the exchange rate; the BCB intervenes only against disorderly conditions, not to set a level.

The nominal anchor moved from the exchange rate to the inflation target.

Source: BCB, Foreign Exchange Policy.

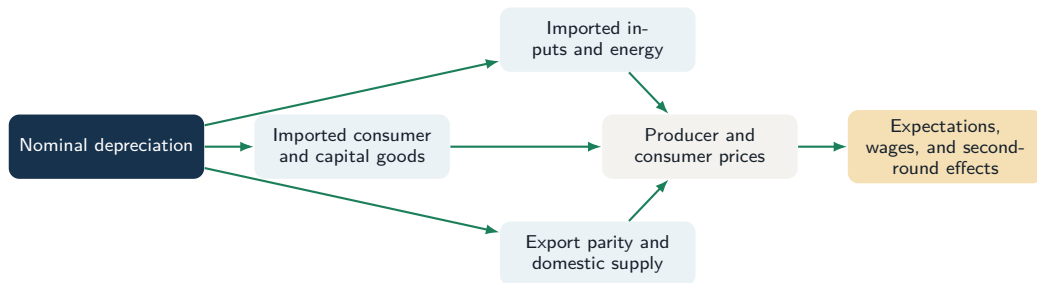
The exchange rate adjusted after the 1999 regime change



- The band limited adjustment before 1999; the float then absorbed domestic and external shocks.
- A weaker real is not a development policy by itself: relative prices, balance sheets, and persistence matter.

Source: BCB SGS 3692. BRL per US dollar, selling rate at year-end; shaded period marks the explicit-band era.

A depreciation reaches inflation through several margins



- ▶ A 2026 BCB exercise: a 10% depreciation adds about 1.0 point to consumer inflation over 12 months — more in crises, when expectations are unanchored, less in calm times.
- ▶ Monetary policy responds to the inflation forecast, not mechanically to the exchange rate.

Source: BCB, March 2026 Monetary Policy Report, pass-through box; BCB Working Papers 5, 181, and 473. Estimates are episode-dependent.

A weaker currency has ambiguous development effects

Who	Possible gain from depreciation	Possible cost
Exporters	Higher domestic-currency revenue	Imported inputs and financing become costlier
Households	Employment may shift toward tradables	Food, fuel, and durable goods become dearer
Productivity	Entry into tradables may support learning	Protection can preserve low-productivity firms

The answer depends on balance sheets, imported-input intensity, credibility, and persistence.

Source: Full five-actor version, including firms and the public sector, in the appendix.

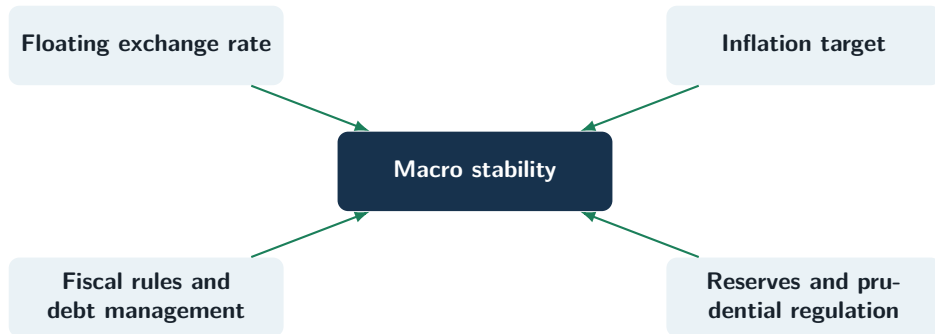
Reserve accumulation changed Brazil's external buffer



- ▶ Reserves insure against external funding stops and reduce foreign-currency liquidity risk.
- ▶ The stock rose sharply before the global financial crisis and has remained large.

Source: BCB SGS 13621. International reserves, cash concept; last daily observation of each month, US\$ billion.

The post-1999 framework combines adjustment and anchors



- ▶ The pieces are complements: fiscal risk affects interest rates and the currency.
- ▶ The float absorbs shocks; monetary credibility limits pass-through.

Brazil's fiscal problem is about composition and credibility

The state is not small

Revenue and social spending are substantial for an upper-middle-income economy.

Much spending is rigid

Pensions, payroll, transfers, and earmarks limit annual reallocation.

Development margin

Public investment fell to roughly 2% of GDP — among the lowest shares in the G20 — as rigid current spending grew.

Macro margin

Uncertain debt paths raise risk premia and shorten private planning horizons.

Source: OECD, [Economic Survey: Brazil 2023](#).

Debt dynamics compress the space for development spending



- ▶ Debt rose before the pandemic, jumped in 2020, and remains high relative to the mid-2000s.
- ▶ Interest rates, primary balances, growth, and debt composition jointly determine the path.

Source: BCB SGS 13762. Gross general government debt, methodology used since 2008, monthly, percent of GDP.

Low saving and investment constrain capital deepening

16.8%
investment / GDP, 2025

14.4%
saving / GDP, 2025

- ▶ Capital per worker grows slowly when investment remains low.
- ▶ High financing costs screen out long-horizon projects.
- ▶ Infrastructure bottlenecks reduce the return to private capital.
- ▶ Productivity growth matters because more investment alone can misallocate resources.

Source: IBGE, National Accounts 2025; rates are shares of GDP.

Macro synthesis: three messages

1. **The Real Plan was an institutional solution**, not just a new currency: the URV coordinated expectations, and fiscal measures backed it.
2. **1999 changed who absorbs shocks**: the float, the inflation target, fiscal rules, and reserves work as one architecture.
3. **Stability is not convergence**: debt uncertainty, rigid budgets, and a high cost of capital still shorten horizons and compress investment.

Stable prices solved one historical constraint. Inequality, weak diffusion, and uneven state capacity remained.

Source: Full macro scorecard table in the appendix.

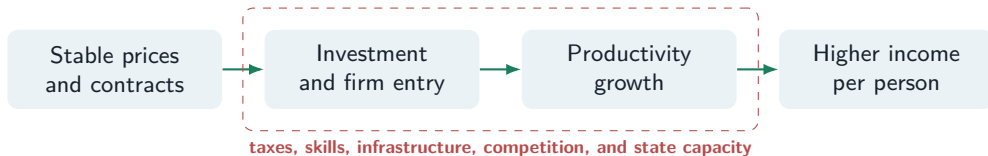
Short break

After the break: how historical formation appears in Brazil today

Part I – Brazil today

Institutions, inequality, production, and state capacity

Stability raises the return to investment; it does not guarantee it



Brazil solved much of the first arrow. The middle of the chain remains weak.

Aggregate productivity depends on firms and allocation

$$\text{Aggregate productivity} = \underbrace{\text{productivity within firms}}_{\substack{\text{technology, management,} \\ \text{skills, scale}}} + \underbrace{\text{allocation across firms}}_{\substack{\text{entry, exit, competition,} \\ \text{capital and labor flows}}}$$

Example: if a bakery that produces twice as much per worker wins customers from a less productive one nearby, average productivity rises — even though no single firm changed.

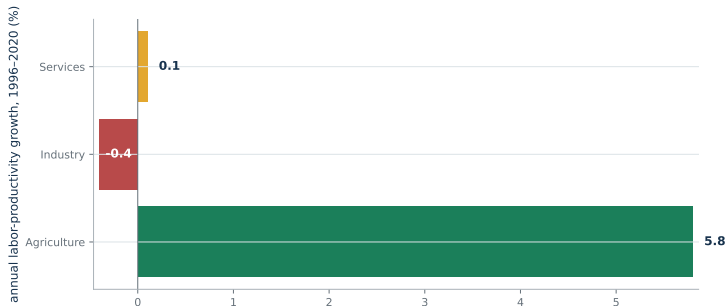
Within-firm margin

Can an existing producer adopt better technology and organization?

Reallocation margin

Do productive firms expand while persistently unproductive firms shrink or exit?

Brazil's productivity problem is not uniform across sectors



- ▶ The bars are average annual growth rates of *labor productivity* by sector, 1996–2020.
- ▶ Agriculture shows that sustained productivity growth is possible; services employ most workers, so diffusion beyond the frontier sector determines the aggregate result.

Source: World Bank, labor-productivity growth by sector, 1996–2020; values rounded.

A long tail of low-productivity firms shapes the aggregate

The long tail

Many small producers use little capital, management technology, or formal finance.

The frontier

A smaller set of firms competes globally and adopts advanced technology.

- ▶ Capabilities decide adoption: targets, inventory control, quality systems, and skills.
- ▶ Complex rules encourage firms to remain small or split activities.
- ▶ Weak exit and reallocation preserve low-productivity uses of capital and labor.

Credit subsidies alone do little when management, competition, and demand remain unchanged.

Source: World Bank, *Brazil: Skills and Jobs* and productivity diagnostics; OECD, *Economic Survey: Brazil 2023*.

Low unemployment can coexist with a dual labor market



Informality

38.1%
of workers in 2025

- insurance: income and flexibility
- trap: no credit, scale, or training
- policy: lower formalization cliffs, portable benefits

Source: BCB SGS 24369, sourced from IBGE PNAD Contínua; IBGE annual informality rate for 2025. Informality is a statistical labor-market category, not a synonym for tax evasion.

State capacity is an economic input

Administrative capacity

Select projects, hire staff, procure inputs, manage contracts, and use data.

Legal capacity

Enforce contracts, protect rights, resolve disputes, and sanction corruption.

Fiscal capacity

Raise revenue with low compliance costs and allocate it toward high-return services.

Coordination capacity

Align federal, state, and municipal responsibilities across shared networks.

Keep this slide in mind: every wedge that follows — taxes, infrastructure, education, security — is also a state-capacity problem.

Source: World Bank, The Brazil of the Future; OECD, Government at a Glance: Latin America and the Caribbean.

Spatial inequality changes which reform binds first

Productive structure

Agriculture, industry, services, and public employment have different regional weights.

Local capacity

Municipalities differ in tax bases, technical staff, procurement, and service networks.

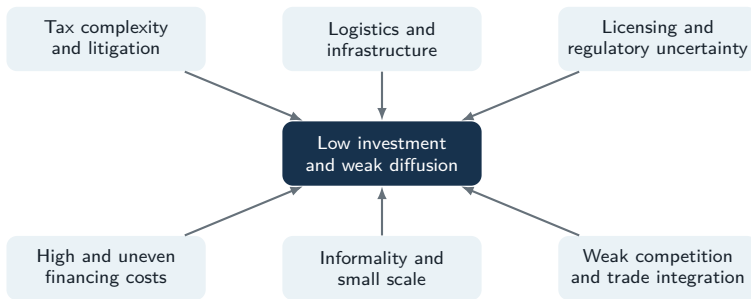
Household access

Transport, schools, health, sanitation, and security shape mobility within the same city.

A national reform produces local results through unequal implementation capacity.

Source: IBGE, Regional Accounts and 2022 Population Census; Ipea, regional development studies.

“Custo Brasil” is a system of interacting wedges



Removing one wedge may have little effect when the other complements remain binding.

The rest of this block walks through these wedges one by one.

Brazil is replacing five overlapping taxes with two VATs

2026	2027	2029–32	2033
Test year for the new system.	Federal VAT (CBS) replaces PIS/Cofins.	State-municipal VAT (IBS) phases in for ICMS/ISS.	Full new system.

- ▶ Today, five taxes at three government levels tax the same sale — taxes pile on taxes (“cascading”) and firms litigate for decades.
- ▶ A destination-based VAT taxes only the value each firm adds, reducing production distortions.
- ▶ Federal coordination, exemptions, compliance systems, and transition rules determine the gain: implementation is an economic reform, not an administrative footnote.

Source: Receita Federal, [Consumption Tax Reform](#).

Competition policy changes the return to innovation

When entry is credible

Incumbents invest, lower markups, and adopt technology to defend market share.

When rents are protected

Licensing, procurement design, network bottlenecks, and local political connections can preserve inefficient firms.

- ▶ Antitrust, bankruptcy, trade exposure, and transparent procurement affect the same reallocation margin.
- ▶ The objective is contestability, not a mechanical preference for large or small firms.

Source: OECD, Competition Assessment Reviews: Brazil; CADE, institutional reports.

Infrastructure gaps: logistics and sanitation

Logistics

- ▶ A continental economy makes transport time and reliability part of marginal cost.
- ▶ Roads, rail, ports, storage, and digital systems are complements.

Sanitation

- ▶ Water and sewage coverage affect disease, school attendance, time use, and property values.
- ▶ National averages conceal large regional and within-city gaps.

A new asset raises productivity only if governance, maintenance, and the network around it work.

Source: World Bank, Brazil infrastructure diagnostics; EPL/Infra S.A., National Logistics Plan; SNIS, Ministry of Cities.

Trade can raise productivity through inputs, scale, and selection

Channels

- ▶ Cheaper and better intermediate inputs
- ▶ Larger markets for productive exporters
- ▶ Technology and management diffusion
- ▶ Entry and exit under competition

Transition costs

- ▶ Workers and places adjust slowly
- ▶ Logistics and customs can bind first
- ▶ A volatile real exchange rate changes incentives
- ▶ Skills determine who captures the gains

The relevant choice is the sequence and support for adjustment, not autarky versus free trade.

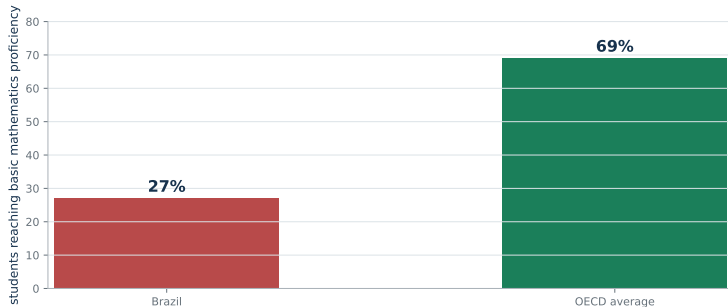
The credit spread remains a microeconomic wedge



- ▶ Concretely: with a policy rate near 10% and a 20-point spread, a small firm borrows at about 30% a year — a machine must return that much before financing it makes sense.
- ▶ The spread combines expected losses, operating costs, taxes, market power, and risk premia; collateral rules, information sharing, competition, and insolvency reform address different parts.

Source: BCB SGS 20783. Average spread on outstanding credit, monthly, percentage points.

The human-capital constraint is learning, not enrollment alone



- ▶ PISA tests 15-year-old students; “basic proficiency” is Level 2 in mathematics — using simple procedures to solve everyday problems.
- ▶ Learning gaps reduce technology adoption and worker mobility.

Source: OECD, [PISA 2022: Brazil](#).

Human capital starts before school and differs by place

Early childhood

Nutrition, health, language exposure, and childcare shape readiness.

School system

Teacher support and instructional time determine whether enrollment becomes learning.

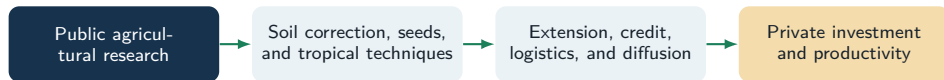
Territory

Transport, violence, sanitation, and local fiscal capacity affect access and continuity.

- ▶ Childcare pays a double dividend: mothers can take formal jobs, and quality care improves children's language, nutrition, and school readiness.
- ▶ Equal spending does not imply equal learning when local constraints differ.

Source: World Bank, Brazil Human Capital Review; IBGE, PNAD Contínua care responsibilities; Inep education indicators.

Embrapa shows how complementary institutions raise productivity



Lesson

Mission-oriented research works when adoption, finance, and infrastructure reinforce it.

Trade-off

Productivity per hectare and environmental enforcement jointly determine land-use outcomes.

Source: Embrapa Cerrados; Ipea, agricultural productivity studies.

Inequality is racial, regional, and gendered

Race and income

White workers earn about 70% more on average; Black and brown workers are overrepresented in informal jobs.

Region and services

Sanitation, transport, and school gaps concentrate in majority-Black peripheries and the North and Northeast.

Gender and care

Women carry more unpaid care work and face lower labor-force participation, especially when childcare is scarce.

- ▶ Black and brown Brazilians are 55.5% of the population. Race, region, and gender overlap in access to safe neighborhoods, schools, formal work, and political voice.

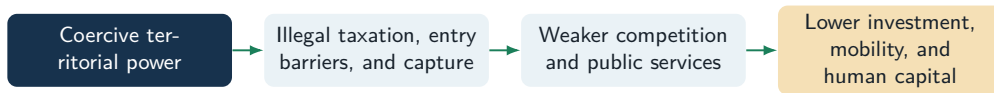
These are not separate topics: they determine who can build skills, move, invest, and participate.

Source: IBGE, 2022 Census, Desigualdades Sociais por Cor ou Raça no Brasil, and PNAD Contínua care and labor-force indicators; Ipea, Poverty and Inequality 1995–2024.

Case study – Security and development

How coercive territorial power changes markets, services, and state capacity

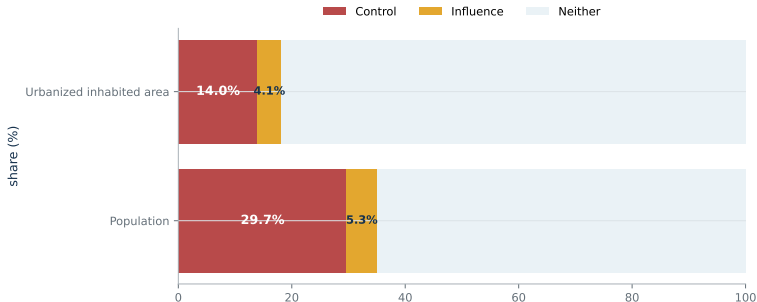
Organized crime is a microeconomic constraint



- ▶ Extortion acts like an illegal tax; forced suppliers create local market power over gas, internet, and transport.
- ▶ Land appropriation weakens property rights and collateral.
- ▶ Intimidation and corruption can redirect enforcement and public resources.

Source: MPRJ documented cases; SENAPPEN; GENI/UFF and Instituto Fogo Cruzado. Mechanisms are not estimates of aggregate GDP losses. Measurement details in the appendix.

Metropolitan Rio shows the scale of territorial armed governance



- ▶ In 2024, 34.9% of residents were estimated to live under armed-group control or influence.
- ▶ The estimate covers Metropolitan Rio, uses anonymous reporting, and is not a national census — this is a Rio case study, not a national statistic.

Source: GENI/UFF and Instituto Fogo Cruzado, Mapa Histórico dos Grupos Armados 2025, with 2022 Census population data. Published shares are rounded.

Territorial violence erodes human capital

Exposure

51%

of students in the 19 municipalities studied were affected by armed violence in 2022.

Learning

10 points

maximum reported SAEB gap, interpreted by the authors as about six months of learning.

Dropout

**7.2 vs.
9.0–9.6%**

non-controlled versus controlled areas in Metropolitan Rio.

Source: UNICEF, Educação Sob Cerco 2025, with GENI/UFF and Instituto Fogo Cruzado. Conditional estimates with statistical controls; the design is not experimental.

A credible response combines enforcement and service delivery

Protect territory and services

Safe routes, reliable transport, service continuity, youth prevention.

Outcome: attendance, business entry, service uptime.

Investigate and prosecute

Forensics, witness protection, case management, corruption control.

Outcome: clearance with due process.

Follow the money

Beneficial ownership, financial intelligence, freezing and confiscation.

Outcome: assets recovered, cash flow disrupted.

Success is measured by market entry and school attendance — not by operations or arrests alone.

Source: MJSP, ENFOC and ENCCLA; SENAPPEN/CNCP, National Plan 2024–2027. Full instrument-outcome table and discussion exercise in the appendix.

Part II – Where do we go from here?

Problems to solve and opportunities to develop

Four structural problems limit the next transition

Organized crime

Territorial control weakens the state, public services, property rights, and market entry.

Fiscal challenges

INSS obligations, rigid spending, high debt, and interest costs compress investment and policy space.

Education

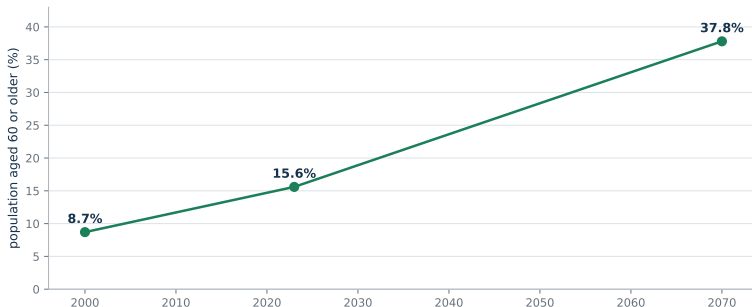
Enrollment has expanded, but weak learning limits productivity and social mobility.

Innovation

Research, firms, finance, competition, and diffusion remain weakly connected outside selected sectors.

Brazil must solve these problems while demography and climate narrow the time available.

Demography raises the return to productivity now



- ▶ The working-age share will no longer provide the same contribution to growth.
- ▶ Health, pensions, care, lifelong learning, and later-life employment become linked policies.

Source: IBGE, Population Projections 2024. The 2070 value is a projection, not an observed outcome.

Three clocks change every block you have seen

Demography

Population peaks around 2041. The 60+ share rises from 15.6% in 2023 to 37.8% in 2070.

Climate

Droughts, floods, heat, and land-use pressure affect capital, food, energy, and cities.

Fiscal space

Aging raises health and pension pressure while adaptation and infrastructure require investment.

Future growth must rely more on productivity because labor-force expansion is ending.

Source: IBGE, Population Projections 2024; OECD, Economic Survey Brazil 2023.

Implementation is part of every reform

The federal bargain

One Union, 27 states, and 5,570 municipalities share taxes and mandates; major reforms need constitutional amendments.

A rigid budget

Earmarks and mandatory spending leave little room to reallocate each year.

Concentrated losers, diffuse winners

Sectors with special regimes organize and litigate; winning consumers are dispersed. The 2023 tax reform passed by compensating losers over decades.

Unequal implementation

A law is only as good as the local administration that applies it.

A reform's design includes its coalition, compensation, and sequencing.

Productive opportunities

Existing strengths can support new industries, exports, and higher-productivity services

Climate is a production strategy, not a final section

Energy advantage

86.6%

renewable domestic electricity supply in 2025

- ▶ Cheap clean power is an export and investment pitch: low-carbon industry, fuels, and agriculture.
- ▶ Climate runs through every earlier block: exports and the exchange rate, infrastructure and adaptation, agriculture, urban risk, and the budget.

Emissions structure

about 70%

from land use and agriculture in the 2022 inventory

- ▶ Enforcement, livestock productivity, and restoration decide credibility abroad.

Source: EPE, BEN 2026; MCTI, National Emissions Inventory, 2022 results.

Brazil can expand four underused advantages

Clean energy

Renewable electricity, biofuels, and low-carbon fuels can attract production and support new exports.

Agribusiness

Research and tropical agriculture can extend into logistics, technology, processing, and higher-value products.

Knowledge cities

São Paulo and Rio combine universities, researchers, firms, capital, and specialized workers.

Tourism

Natural, cultural, and urban attractions remain constrained by safety, connectivity, and local public services.

The opportunity is not a blank slate: it starts from capabilities Brazil already has.

Established capabilities can seed new global industries

Established global industries

- ▶ Mining — Vale
- ▶ Oil and energy — Petrobras
- ▶ Pulp and paper — Suzano and Klabin

Emerging opportunities

- ▶ Climate-related industries
- ▶ Digital products and services
- ▶ Financial technology — Pix

The development task is to connect existing firms, research, finance, workers, and suppliers.

Pix shows that Brazilian institutions can build at scale

170m+
individual Pix users

7bn+
transactions in January 2026

- ▶ The central bank created a free, instant-payment platform that reached national scale within a few years.
- ▶ Pix complements private banks and fintech firms; common infrastructure expands entry and innovation.

The lesson: capable public institutions can create markets rather than merely regulate them.

Source: BCB, [Pix Statistics](#); figures reported in early 2026.

The agenda joins structural problems to productive capabilities

Historical formation → Brazil today

Structural problems

Fiscal pressure, organized crime, weak learning, limited innovation, and unequal access to state capacity.

Development task

Solve fundamental institutional problems.

Productive capabilities

Clean energy, agriculture, knowledge cities, global firms, natural resources, and digital finance.

Development task

Expand and connect existing capabilities.

Both tasks lead to the same goal: a richer and more productive economy.

Brazil must solve constraints and expand capabilities together

What would be required to complete the transition?

Brazil already has substantial productive and institutional assets. The central challenge is to combine them with progress in fiscal policy, public security, education, and innovation.

Solve

Restore fiscal space, territorial control, learning, and the institutional links that turn research into innovation.

Expand

Use clean energy, productive agriculture, urban knowledge, global firms, tourism, and digital finance as platforms for diffusion.

Appendix

Backup slides: formulas, full tables, measurement, and references

Backup: nominal and real exchange rates answer different questions

Nominal rate

$$E = \text{BRL per USD}$$

A rise in E is a nominal depreciation of the real.

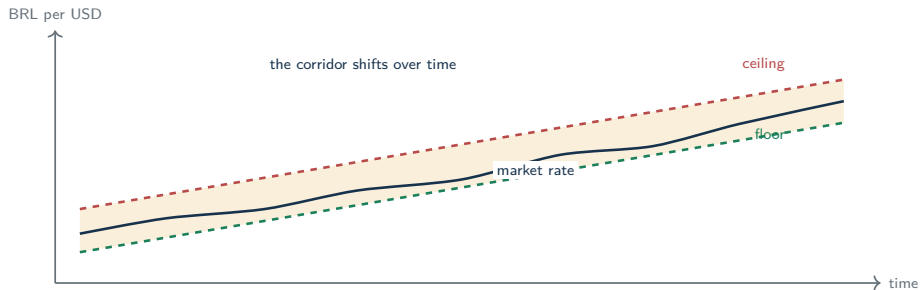
Real exchange rate

$$q = \frac{EP^*}{P}$$

A rise in q makes foreign goods more expensive relative to domestic goods under this definition.

- ▶ Domestic and foreign inflation can undo a nominal depreciation.
- ▶ Productivity, terms of trade, and risk premia affect the equilibrium real rate.
- ▶ “Competitive” has no meaning without a price index, horizon, and sector.

Backup: a crawling band trades flexibility for credibility



- The market rate fluctuates inside the corridor while the authorities move its boundaries over time.
- A narrow corridor can strengthen the anchor but invites one-way bets when fundamentals diverge.

Source: Stylized diagram; BCB, 1999 Q1 FX Market Analysis.

Backup: monetary policy reacts to the inflation outlook



- ▶ The policy rate is forward-looking; it need not move one-for-one with current inflation.
- ▶ Fiscal risk and exchange-rate pass-through affect how much tightening is required.

Source: BCB SGS 432 and 433. Selic target is the last observation of each month; IPCA is compounded over 12 months.

Backup: full distributional effects of a depreciation

Margin	Possible gain from depreciation	Possible cost
Exporters	Higher domestic-currency revenue	Imported inputs and financing become costlier
Import-competing firms	More price space	Capital goods and technology cost more
Households	Employment may shift toward tradables	Food, fuel, and durable goods become dearer
Public sector	Some revenues rise with nominal activity	Indexed debt and inflation costs can increase
Productivity	Entry into tradables may support learning	Protection can preserve low-productivity firms

The answer depends on balance sheets, imported-input intensity, credibility, and persistence.

Backup: macro scorecard

Area	Achievement	Remaining constraint
Inflation	Hyperinflation ended; expectations have an explicit anchor	Shocks still pass through the currency and indexation
Exchange rate	A float absorbs external shocks	Volatility affects prices, balance sheets, and investment
External sector	Reserves and lower FX mismatch strengthened buffers	Commodity and global-rate cycles remain important
Fiscal policy	Strong revenue capacity and broad social insurance	Rigid spending and debt uncertainty compress investment
Capital formation	Deep domestic financial markets	High cost of capital and low investment

Backup: micro scorecard

Area	Evidence of capability	Binding margin
Technology	Agricultural research and digital finance	Diffusion beyond frontier sectors
Firms	Globally competitive exporters and banks	Long tail of small, informal producers
Tax system	Reform establishes a modern VAT structure	Transition, exemptions, and implementation
Trade	Competitive commodity and selected manufacturing exports	Low integration and costly inputs
Workers	Broad enrollment and recent labor-market strength	Learning gaps and informality
Infrastructure	Scalable concessions and digital public rails	Logistics, sanitation, and local coordination
State capacity	Strong agencies and data systems in selected areas	Uneven local capacity and territorial coercion

Backup: there is no single organized-crime statistic

Lethal violence

Unit: deaths

National and annual. It does not identify the organization responsible.

Prison intelligence

Unit: groups

SENAPPEN identified at least 88 organizations affecting prisons in its 2024 exercise.

Territorial control

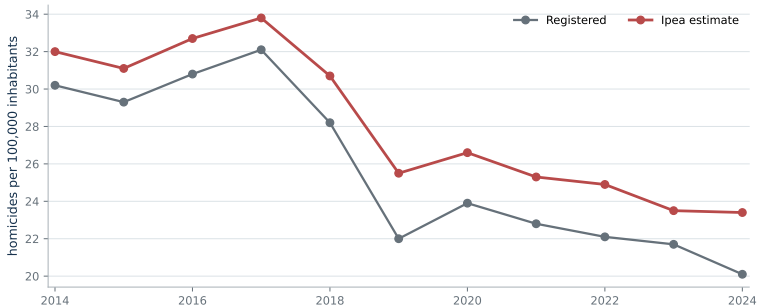
Unit: places and residents

Directly measures armed governance, but only for Metropolitan Rio.

Measurement must match the mechanism being discussed.

Source: Ipea/FBSP, Atlas da Violência 2026; SENAPPEN, Mapa das ORCRIMs; GENI/UFF and Instituto Fogo Cruzado, 2025.

Backup: lethal violence fell, but data quality changes the trend



- ▶ The registered rate fell after 2017; reallocating some deaths of undetermined cause produces a smaller decline.
- ▶ Homicide measures lethal violence, not the size or revenue of criminal organizations.

Source: Ipea and FBSP, Atlas da Violência 2026, based on SIM/Ministry of Health and IBGE. Rates per 100,000. Estimated series is model-based.

Backup: coercive market power reaches ordinary goods and services

Household services

Cooking gas, broadband, cable TV, and electricity.

Mobility

Alternative transport and compulsory payments from drivers.

Property

Irregular construction, forced dispossession, and property sales.

Local retail

Security fees, merchant extortion, and forced suppliers.

Economic wedges

- ▶ illegal tax
- ▶ restricted entry
- ▶ weak property rights
- ▶ distorted procurement

Source: MPRJ cases in Nova Iguaçu (2019), Itaboraí (2019), Queimados (2024), and the illegal-construction task force (2022). These cases do not measure national prevalence.

Backup: full instrument-outcome menu for security policy

Margin	Instruments	Development outcome
Territorial prevention	Safe routes, reliable transport, service continuity, youth prevention	Attendance, business entry, service uptime
Investigation and justice	Forensics, witness protection, case management, corruption control	Clearance with due process, time to disposition
Financial disruption	Beneficial ownership, intelligence, freezing and confiscation	Assets finally recovered, disrupted cash flow
Sector regulation	Internet, fuels, fintech, construction, health, procurement	Prices, entry, ownership transparency
Prison governance	Intelligence, lawful communications control, work and reintegration	Violence, illicit communication, recidivism

Source: MJSP, ENFOC and ENCCLA; SENAPPEN/CNCP, National Criminal and Penitentiary Policy Plan 2024–2027.

Backup discussion: what outcome would demonstrate greater state capacity?

Choose one instrument and the final outcome that should be used to evaluate it.

Operations

Activity measure

Arrests

Intermediate output

Recovered assets

Financial outcome

Market entry

Development outcome

- ▶ Add one safeguard against abuse or displacement to another territory.
- ▶ Explain why your chosen outcome is closer to welfare than the activity count.

References and data

Data cutoff: July 2026. Series retrieved from official APIs on July 21, 2026; processed CSVs archived with the slides.

Time series (BCB SGS codes)

- ▶ IPCA, monthly: SGS 433
- ▶ Selic target: SGS 432
- ▶ BRL/USD, year-end selling rate: SGS 3692
- ▶ Real GDP growth (IBGE): SGS 7326
- ▶ International reserves, cash: SGS 13621
- ▶ Gross general government debt: SGS 13762
- ▶ Unemployment (PNAD Contínua): SGS 24369
- ▶ Credit spread, outstanding: SGS 20783
- ▶ Convergence: World Bank WDI, NY.GDP.PCAP.KD (constant 2015 US\$)

Reports and studies

- ▶ BCB, Monetary Policy Report, March 2026 (pass-through box); Working Papers 5, 25, 181, 473
- ▶ OECD, Economic Survey: Brazil 2023; PISA 2022 Brazil country note
- ▶ World Bank, The Brazil of the Future (productivity, 1996–2020)
- ▶ IBGE: 2022 Census; Regional Accounts 2021; Population Projections 2024; Desigualdades Sociais por Cor ou Raça
- ▶ Ipea/FBSP, Atlas da Violência 2026
- ▶ GENI/UFF and Instituto Fogo Cruzado, Mapa Histórico dos Grupos Armados 2025
- ▶ UNICEF, Educação Sob Cerco 2025
- ▶ EPE, BEN 2026; MCTI, National Emissions Inventory (2022)
- ▶ Receita Federal, Consumption Tax Reform portal